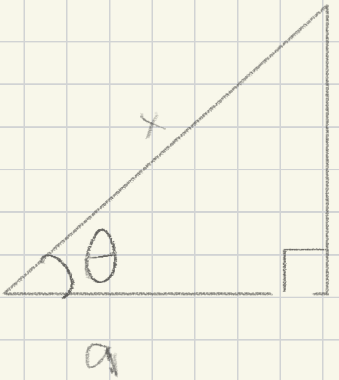
Sep 15 warmup

What trig substitution is used to transform the integral into a trig integral

$$\int \frac{1}{x^2-a^2}\,dx \quad (a>0)$$



$b = \sqrt{x^2-a^2}$

$\sin\theta = \frac{b}{x} = \frac{\sqrt{x^2-a^2}}{x}$

$\cos\theta = \frac{a}{x} \to x = a\sec\theta$

$\tan\theta = \frac{b}{a} = \frac{\sqrt{x^2-a^2}}{a}$

$\hookrightarrow a\tan\theta = b$

$a^2\tan^2\theta = b^2$
$= x^2-a^2$

$x = a\sec\theta \quad dx = a\sec\theta\tan\theta\,d\theta$

$\int \frac{1}{a^2\tan^2\theta}\, a\sec\theta\tan\theta\,d\theta$

$= \int \frac{\cancel{a}\sec\theta\cancel{\tan\theta}}{a^{2-1}\tan^{2-1}\theta}\,d\theta$

$= \int \frac{\sec\theta}{a\tan\theta}\,d\theta = \int \frac{1}{a}\frac{\sec\theta}{\tan\theta}\,d\theta$

$= \frac{1}{a}\int \frac{1}{\cos\theta}\cdot\frac{1}{\tan\theta}\,d\theta$

$= \frac{1}{a}\int \frac{1}{\cancel{\cos\theta}}\cdot\frac{\cancel{\cos\theta}}{\sin\theta}\,d\theta = \frac{1}{a}\int\frac{1}{\sin\theta}\,d\theta$

$= \frac{1}{a}\int\csc\theta\,d\theta$

But this would be complex to integrate, hence why we'd use partial fractions integration.